Application Detail

Application ID	2249716
Submitted	Mar 9, 2026
Status	In progress
Applicant(s)	David Molik (david.molik@gmail.com)
Organization	KANSAS STATE UNIVERSITY 48-0771751 105 ANDERSON HALL Suite 200, MANHATTAN, KS, 665060100, US
Program and cycle	2026 New Innovator Award Nomination Always Open
Tags	No tags
Forms	2026 New Innovator Award - Full Application

Full Proposal Details

2026 New Innovator in Food and Agriculture Research Award

Unique Circumstances: 2026 Awards Contingent on Funding Availability

FFAR's model requires federal funding from Congress to build public-private partnerships. Due to delays in passing a new Farm Bill, FFAR's funding is limited and all remaining program funds are currently committed and expected to be awarded in 2026. FFAR's ability to fund research is contingent on receiving additional funding. The 2026 New Innovator Award RFA, which would typically result in 2027 awards, is subject to this limitation.

FFAR wishes to advise potential applicants that it cannot fund any New Innovator Awards selected through this RFA until additional funds are allocated to the Foundation.

FFAR is proceeding with this RFA to ensure that the Foundation can quickly fund new research should funds become available. The award of any project selected from this RFA is contingent on the availability of funds.

FFAR strongly encourages potential applicants to consider our current funding situation when deciding whether to apply for the 2026 New Innovator Award.

Key Dates

Nomination Open: February 11, 2026 at 12:00 pm EST
Nominations Due: March 18, 2026 by 5:00 pm EDT

NOTE: FFAR will review nominations against eligibility criteria. Only those meeting these criteria will be invited via email to submit full applications.

Full Application Invitation: March 25, 2026
Full Applications Due: April 29, 2026 by 5:00 pm EDT

Objectives

The Foundation for Food & Agriculture Research (FFAR) seeks nominations for its 2026 New Innovator in Food and Agriculture Research Award. Through this program, FFAR intends to provide early-career scientists with the investment needed to propel them into successful research careers. The New Innovator Award program supports and promotes exceptionally talented new faculty who are conducting critical and highly innovative research addressing significant food and agriculture challenges. These awards allow grantees to focus exclusively on research without the pressure of securing additional funding.

The New Innovator Award is a flagship program of FFAR's Scientific Workforce Priority Area and seeks to promote career advancement of promising new scientists who intend to make a long-term career commitment to research in food and agriculture and bring innovative, ground-breaking research initiatives and thinking to bear on problems facing food and agriculture. Within the scope of the New Innovator Award program, investigators have the freedom to explore new avenues of inquiry that arise during their research. Therefore, FFAR is interested in the program of research to be explored and its impact as opposed to a list of very specific aims.

It is expected that, through this program, investigators will have the ability to pursue highly creative ideas, firmly establish their early lines of research into important food and agriculture scientific areas and provide funding stability in the critical early career years.

Before You Begin

Helpful Tips

- Use Manage Applicants above to add any additional individuals that should have access to this form and any subsequent forms related to this application.
- Add mail@grantapplication.com and noreply@yourcause.com to your address book or safe senders list to ensure you receive all system communications.
- Before filling out the application, you may find it helpful to click the Download icon above. This will offer you a preview of all the questions on the application.
- You may wish to utilize a word-processing tool (i.e. Microsoft Word), then cut and paste your responses into the application.
- Your application will be saved automatically. If you want to return later, simply close the browser. You may return by going to: https://bbgmapply.yourcausegrants.com/apply/applications
- Additional help can be found on the FFAR's website at https://foundationfar.org/grants-funding.

Organization Information

Organization Information

Organization Name*

Kansas State University

Organization Type*

Non-Profit/College/University

Is the applicant organization a Startup company?*

No

For the purposes of this application, FFAR defines a Startup as an entity that has at least one of the following descriptors:

- An independently owned or privately owned entity,
- Newly emerging company typically incorporated for less than 5-10 years,
- Uses investments from venture capital companies, private equity firms, family or friends, angel investors, crowd-sourcing, or Small Business Innovation Research (SBIR) or other grants rather than generating revenue from sales, consulting, etc.

Tax ID*

Required for all US applicants. Please include numbers and dashes.If organization is international, please type "None."

48-0771751

DUNS Number

Required for all US applicants. Please include numbers and dashes.

92-977-3554

Congressional District*

Required for all US applicants. Please include numbers and dashes. If not in the US, please type "International."

KS-001

Address*

Please provide the geographic location of the applicant's organization/institution.

1601 Vattier Street, Manhattan, KS 66506, USA

Other Country
If "Other" was selected, please provide the name of the country.

No answer

Phone*

785-532-6804

Website Address

https://www.k-state.edu/

Nominator/Authorized Organizational Representative (AOR)

The Authorized Organization Representative (AOR) is the administrative official who, on behalf of the applying organization, is authorized to make certifications, assurances, and enter contracts and agreements on behalf of the applicant to fulfill the terms and conditions of the grant award that is being submitted to FFAR.

Contact Information*

Please adhere to the following character limits:
First Name - 100 characters
Last Name - 100 characters
E-Mail - 75 characters
Phone - 30 characters
Institution - 255 characters
Minimum number of responses: 1
Records: 1

Prefix:	Mr.
First name:	Paul
Last name:	Lowe
Suffix:	
Institutional E-mail:	research@ksu.edu
Phone:	785-532-6804
Office address:	1601 Vattier Street, Manhattan, KS, 65506, US
Institution:	Kansas State University
Contact Title:	Associate Vice President for Research

Grant Administrator

Contact Information*
Please adhere to the following character limits:
First Name - 100 characters
Last Name - 100 characters
E-Mail - 75 characters
Phone - 30 characters
Institution - 255 characters
Minimum number of responses: 1
Records: 1

Prefix:	Ms.
First name:	Nidhi
Last name:	Bhandari
Suffix:	
Institutional E-mail:	nidhim@ksu.edu
Phone:	785-532-6804
Office address:	1601 Vattier St 103 Fairchild Hall, Manhattan, KS, 66506, US
Institution:	Kansas State University
Contact Title:	Lead Grant Specialist

Press Contact

Contact Information*
Please adhere to the following character limits:
First Name - 100 characters
Last Name - 100 characters
E-Mail - 75 characters
Phone - 30 characters
Institution - 255 charactersters
Minimum number of responses: 1
Records: 1

Prefix:	Ms.
First name:	Erin
Last name:	Pennington
Suffix:	
Institutional E-mail:	Erin120@k-state.edu
Phone:	785-532-5045
Office address:	119 Fairchild Hall Kansas State University, Manhattan, KS, 66506, US
Institution:	Kansas State University
Contact Title:	Director of Strategic Research Communications

Governmental/Legislative Contact

While we cannot lobby, FFAR works with Congress and federal agencies to identify research gaps, share results and communicate our value to taxpayers. We hope to coordinate with your institution's Government Relations contact to highlight your research.

If a contact is not available for this category, please enter "None" in the Last Name field.

Contact Information
Please adhere to the following character limits:
First Name - 100 characters
Last Name - 100 characters
E-Mail - 75 characters
Phone - 30 characters
Institution - 255 characters
Records: 1

Prefix:	Mr.
First name:	Paul
Last name:	Lowe
Suffix:	
Institutional E-mail:	research@ksu.edu
Phone:	
Office address:	1601 Vattier Street, Manhattan, KS 66506, USA
Institution:	Kansas State University
Contact Title:	Associate Vice President for Research

Technology Transfer Office Contact

The technology transfer contact is the administrative contact who, on behalf of the applying organization, is authorized to facilitate research collaborations, licensing, and commercialization of new inventions should the Project require it.

Contact Information

Please adhere to the following character limits:
First Name - 100 characters
Last Name - 100 characters
E-Mail - 75 characters
Phone - 30 characters
Institution - 255 characters
Records: 1

Prefix:	Mr.
First name:	Christopher
Last name:	Brandt
Suffix:	Esq.
Institutional E-mail:	brandtcd@ksu.edu
Phone:	785-532-3919
Office address:	
Institution:	Kansas State University
Contact Title:	Associate vice president Kansas State University Research Foundation

Contact Information

Principal Investigator (Nominee)

The PI is the individual who is responsible for the scientific or technical direction of the program and have primary responsibility for the proposed work and reporting requirements, if awarded.

Please provide the details of the PI on this program.

Primary Contact/Principal Investigator*

The nominated faculty member must have been hired on or after August 1, 2022 for a tenture-track or equivalent position and may work in any discipline or any department within the institution.

Preference will be given to individuals near the onset of their independent research career and who are within eight (8) years of receiving a Ph.D. or equivalent degree. ,

Please adhere to the following character limits
ORCID ID - 25 characters
First Name - 50 characters
Last Name - 50 characters
E-Mail - 75 characters
Phone - 30 characters
Extension - 30 characters
Ethnicity (if self-identifying) - 50 characters
Gender (if self-identifying) - 50 characters
Minimum number of responses: 1
Records: 1

Date of Hire:	Feb 22, 2026
Date PH.D or equivalent degree was obtained:	May 20, 2021
ORCID ID:	0000-0003-3192-6538
Prefix:	Mr.
First name:	David
Last name:	Molik
Suffix:	Ph.D.
Institutional E-mail:	molik@ksu.edu
Office Phone:	(785) 532-5399
Extension:	
Office address:	1117 Mid Campus Dr N, Manhattan, KS 66506, USA
Job Title:	Department Head Open Publishing Exchange Associate ProfessorLibrarian
Ethnicity:	White

Ethnicity (if self-identifying, please specify):	
Gender:	Man
Gender (if self-identifying, please specify):	

Key Personnel

Co-Investigators are not allowed for this program.

Key Personnel are individuals who share the responsibility of the scientific or technical direction of the program or contribute to the intellectual design or execution of the program in a substantive, measurable way.

Please list the name(s) and expertise of the program's Key Personnel on the sections below. Any other Key Personnel should be included under the Additional Key Personnel attachment as listed at the end of this application.

If the PI is the only program personnel, please type "0" (zero) in the question below.

Other than the Principal Investigator, how many total Key Personnel will be involved in the proposed program? *

0

Key Personnel
Please add up to 3 Key Personnel, if any.
Please adhere to the following character limits:
First Name - 50 characters
Last Name - 50 characters
E-Mail - 75 characters
Institution - 255 characters
Phone - 30 characters
Records: 0

Not counting the PI/PD, please provide the name, institution and role of the top five key personnel contributing to this project. If not applicable, please enter "N/A" in the Key Personnel Name field.

(1) Key Personnel Name	Institution	Role on this Project
No answer	No answer	No answer

(2) Key Personnel Name	Institution	Role on this Project
No answer	No answer	No answer

(3) Key Personnel Name	Institution	Role on this Project
No answer	No answer	No answer

(4) Key Personnel Name	Institution	Role on this Project
No answer	No answer	No answer

(5) Key Personnel Name	Institution	Role on this Project
No answer	No answer	No answer

Current and Pending Support

Please provide a Current and Pending Support document on the Attachments section before submission. This document should list all ongoing and upcoming projects for the past three (3) years for the PI or PD and any all included key personnel named in the application. A template can be found on FFAR's Applicant Forms & Examples website.

Additional Key Personnel

Please provide a table including the name, institution, expertise, role, email, and phone number of any other Co-Investigators and/or Key Personnel in addition to the three individuals listed above. Your PDF document can be uploaded on the Attachments section before submission.

Proposal Information

Funding Opportunity

How did you hear about this funding opportunity? Please select all that apply.*

Word of Mouth

If Scientific Society or Association Newsletter was selected, please list the source:

No answer

If News Outlet was selected, please list the source:

No answer

If Other was selected, please describe:

Colleague

Program Title

Provide a short title that captures the essence and communicates the importance of the proposed program to a lay reader. *

Unlocking Agricultural Research Data for Better Farming Decisions

Priority Areas

FFAR's Research Strategy focuses on interrelated, strategic research priorities as we strive toward our goal: A U.S. food and agriculture system that supports producers and their communities, sustains the environment and enables processors and retailers to deliver foods that equitably nourish our population. *

In line with the new Research Strategy, FFAR has transitioned from the previous Challenge Area structure to focus on overarching Research Priorities to foster greater collaboration across disciplines. These priorities reflect a systems approach— recognizing the complex relationships between primary production systems, agroecosystems and the food system as a whole. Collectively, these priorities drive transformational systemic change by taking a broad view of U.S. agriculture, considering its social, environmental and economic benefits as well as impacts. We ask that applicants also address how their proposed research program will address at least one of FFAR's Research Priority Areas. The proposed research program should also be expected to clearly benefit the U.S. food and agriculture system.

- **Cultivating Thriving Production Systems: Production systems thrive when input providers and producers have the information and technologies (novel germplasm, digital tools, pest and disease solutions, etc.) to manage their enterprises efficiently and profitably—which includes selecting animals, crops and management practices that support site-specific environmental, social and business goals.**
- **Sustaining Vibrant Agroecosystems: Tools, technologies and integrative models designed at scales needed for site-specific decision making allow producers to create, build and maintain healthy, productive agroecosystems that meet consumer expectations under changing climate conditions. Such solutions must provide flexibility to deal with future uncertainty, and clarify trends and trade-offs in environmental regeneration, productivity and resilience.**
- **Bolstering Healthy Food Systems: In a healthy food system, food processors, distributors and retailers have the information and technologies needed to produce nutritious, environmentally sustainable and culturally appropriate food for consumers. Likewise, consumers require convenient access to safe, nutritious, desirable foods with a balance of macronutrients, micronutrients and protective phytochemicals.**

Select all that apply.

Sustaining Vibrant Agroecosystems

Key Topics

Key topics or areas of research to be covered in the nominee's program (up to 50 words). *

Agricultural data integration, interoperable metadata standards, research data infrastructure, cross-project data synthesis, agroecosystem modeling, and analytical systems that enable reuse of agricultural research data to support decision-making in crop production, pest management, and environmental resilience.

Public Abstract

Should the proposed program be selected for funding, this abstract would be used in the funding announcement, and it is intended for a general audience (do not include proprietary information in this section). Describe, in simple, nontechnical terms, the overall goals of the proposed program, the potential significance of the results, and the impact of the work on advancing the food and agriculture sector. *

Program abstracts must be included only in the online portion of the application and must not be included as a separate attachment. The Grant Management Site cannot support special characters, and any non-ASCII symbols will be converted to question marks ("?") upon submission. The full term instead of the symbol must be used (up to 500 words).

This project is about making agricultural research data easier to use. It focuses on improving how information from studies on crops, soils, pests, weather, and genetics is organized so that it can be more easily combined and understood. Today, agricultural research produces large amounts of valuable data, but much of it is difficult to use. Information is often spread across many different studies, stored in different formats, and not designed to work together. Because of this, important findings are often overlooked or take a long time to bring together, slowing the development of practical solutions for farmers and agricultural professionals. This project will develop new ways to organize research information so that computer algorithms can help connect and compare results from many different studies. It does this by improving the digital labels (called DOIs) that are already used to identify research papers and datasets. These labels will be expanded to include clear, structured information about what was studied and how the data are organized, making it easier for computers to find and combine related results.

The research team will test this approach using real agricultural data, such as crop trials conducted in different locations and studies on pest management. By organizing this information in a more consistent and structured way, the system will allow researchers to quickly identify relevant studies, compare results, and combine findings from multiple sources. This will help scientists, extension specialists, and agricultural professionals make better use of existing research. For example, it can support faster updates to farming recommendations based on results from many field trials across different regions and conditions. In the long term, this project will increase the value of agricultural research by making data easier to find, combine, and use. By improving how research results are connected and applied, it will support better decision-making and help strengthen the productivity and resilience of U.S. agriculture.

Duration

Start and end dates are contingent on funding availability.

Program Start Date*

Jan 1, 2027

Program End Date*

Dec 31, 2029

Total Program Duration (in months) *

36

Geographic Location

City*

Manhattan

State*

KS

Congressional District*

Required for all US applicants. If not in the US, please type "International."

KS-001

Additional Geographic Locations where the proposed research will be conducted.

No answer

USDA Involvement

Is the applicant or any of its matching partners, if applicable, directly associated with the United States Department of Agriculture (USDA)? *

No

Nomination Information

Text of nomination information for reference.

Agricultural research generates vast and diverse datasets spanning: environmental observations; crop and pest dynamics; and genomics, genetics, and breeding (GGB). However, these datasets are typically produced within siloed projects and stored across fragmented repositories and locations using inconsistent metadata standards. As a result, valuable agricultural research data remain difficult to discover, integrate and reuse across studies, and especially difficult for research and extension communities to translate into practical solutions. This fragmentation limits the ability of researchers and decision-makers to synthesize knowledge across experiments, build integrative agroecosystem models and convert research investments into actionable insights for agricultural systems.

The scientific challenge addressed in this project is the lack of interoperable metadata frameworks and analytical infrastructure capable of supporting cross-project synthesis of agricultural research data. While major investments have been made in domain-specific databases and data repositories, most agricultural datasets remain isolated at the project level, lacking the structured metadata and integration pathways necessary for large-scale analysis. This gap represents a significant barrier to developing the integrative models and decision-support tools needed to understand complex agroecosystems under changing environmental and economic conditions.

This project will develop data infrastructure designed to enable agricultural research data to be systematically integrated, analyzed and reused. The research program focuses on three complementary innovations. First, we will develop interoperable metadata frameworks that standardize the description of agricultural research datasets across environmental, genomic and management domains. Second, we will design repository-linked infrastructure that connects institutional and domain repositories through shared metadata schemas and automated data ingestion workflows. Lastly, we will build integrative analytical systems that enable researchers to perform cross-project analyses and generate ecosystem-level insights from previously isolated datasets. In effect, the solution to this challenge is not solely improved data storage, but the creation of connective infrastructure between platforms, projects and scientific domains.

By enabling synthesis and reuse across diverse studies and experimental systems, the proposed infrastructure will support the development of more robust agroecosystem models, improve understanding of system-level dynamics and strengthen the scientific foundation for agricultural decision-making. By enabling agricultural datasets to be integrated and analyzed across experiments, locations and production systems, this work will help translate research investments into practical knowledge that supports producers, researchers and decision-makers. Ultimately, this work will increase the return on public and private research investments by ensuring that agricultural data can be reused to address emerging challenges in crop productivity, pest management and environmental resilience across U.S. agriculture.

Stakeholders

Describe stakeholder groups to be served by the proposed project such as farmers/producers, policy makers, researchers, consumers, the public, industry, etc. (up to 500 words). *

This program serves interconnected stakeholder groups spanning agricultural research, data infrastructure, and applied decision-making, with both a strong institutional foundation and broad national relevance. While the primary development environment is research- and data-infrastructure-focused, this structure is intentional: it addresses upstream limitations in data integration that currently constrain downstream impact for extension systems, advisors, and producers. At its core, the project is anchored within the Institute for Digital Agriculture and Advanced Analytics (ID3A) at Kansas State University, an interdisciplinary environment connecting agronomy, data science, engineering, and extension. ID3A researchers represent an ideal early adopter community, as they routinely generate and analyze heterogeneous datasets across crop systems, environmental monitoring, and precision agriculture. Collaboration within ID3A ensures the framework is developed and tested in real-world research contexts.

Beyond Kansas State University, the program engages regional and national research networks conducting multi-location field trials, long-term agroecosystem experiments, and genomics-enabled breeding programs. These groups generate large volumes of data that are difficult to integrate across studies due to inconsistent metadata and data structures. By enabling machine-actionable, PID-linked representations of scientific findings and data organization, the proposed framework reduces the time and effort required for cross-study synthesis and collaborative analysis. This is particularly important for addressing geographically distributed challenges such as climate variability, pest dynamics, and soil health management across U.S. production systems.

A second stakeholder group includes data repository managers, publishers, and infrastructure providers who play a central role in disseminating agricultural research but currently lack mechanisms to support computational reuse of published results. This project engages this community through direct integration with publishing and metadata infrastructure at Kansas State University, including through the Open Publishing Exchange and New Prairie Press (both led by the PI). These resources provide an operational environment for embedding machine-actionable metadata into publication workflows, enabling a transition from static dissemination toward interoperable, computationally enabled research outputs and increasing the long-term value of the research they steward.
The program also serves extension specialists, crop advisors, and decision-support developers, who operate at the stage of the research data lifecycle where findings are translated into actionable recommendations. These stakeholders must synthesize results across multiple studies, which often requires manual extraction and harmonization of data from numerous publications. By enabling machine-actionable DOI-linked metadata, the proposed system allows relevant studies to be identified, compared, and aggregated programmatically. This reduces the time required to move from research output to applied insight and supports more timely, evidence-based recommendations for crop management, pest control, and environmental stewardship.

Finally, this work benefits the broader agricultural research community and funding organizations, including USDA and FFAR, by increasing the return on investment in agricultural science. By transforming research outputs into interoperable, machine-actionable knowledge objects, the project enables previously disconnected datasets to be reused at scale. The framework will be openly disseminated and designed for integration with existing PID infrastructure, supporting adoption across institutions and contributing to a more connected, computationally enabled agricultural research ecosystem.

Program Description

The description of the scientific plan should be written at a level appropriate for reviewers who are broadly knowledgeable but may not be directly involved in the proposed area of research. The application should focus on the overarching goals of the research program and address the Review Criteria. It is not necessary to provide the proposed research as a series of specific aims. Do not provide a detailed experimental plan. Preliminary data are allowed but not required.

Describe the scientific challenge or opportunity that you propose to address, its importance, how* anticipated outreach and outcomes would have a major impact on a broad area of research relevant to the FFAR Priority Area and how it will create deliverables which impact U.S. Agriculture.

Provide the scientific context by summarizing the current landscape of the field and how your research, if successful, would extend substantially beyond current boundaries.

Describe the overall approach to be taken. A detailed experimental plan and extensive preliminary data are not required, but the applicant will need to clearly provide a description of the program being proposed.

Applicants must also address how the research will be performed in a robust and rigorous manner, including addressing risks and challenges. If not providing a detailed experimental plan or extensive preliminary data, state prominently that this is done in compliance with the RFA (up to 1,000 words).

Agricultural research generates vast and heterogeneous datasets spanning environmental observations, crop and pest dynamics, and genomics, genetics, and breeding (GGB). These data are foundational for understanding and managing agroecosystems, yet they remain fragmented across publications, repositories, and institutions. Despite significant investments in data infrastructure and widespread adoption of FAIR (Findable, Accessible, Interoperable, Reusable) principles (Wilkinson et al., 2016), most agricultural data are not interoperable at the level required for computational synthesis across studies. As a result, integrating knowledge across experiments, production systems, and environments remains slow and labor-intensive, limiting the development of predictive agroecosystem models and timely decision-support tools for U.S. agriculture, and reducing the overall return on public and private research investments.

Stakeholders rely on synthesizing results across multiple studies (e.g., field trials, pest management experiments, and environmental observations) to develop regionally relevant guidance for producers. In current workflows, this requires manual identification, extraction, and harmonization of results from individual publications and datasets, often across dozens or hundreds of studies (Haddaway et al., 2020). Although research outputs are linked through Digital Object Identifiers (DOIs), which include structured metadata schemas (e.g., Crossref and DataCite), these are designed primarily for discovery and citation rather than encoding scientific meaning or data structure (Fenner et al., 2019). As a result, synthesis remains time-intensive, difficult to reproduce, and slow to incorporate new data.

This project addresses this limitation by enabling machine-actionable, PID-linked representations of scientific findings and data structure. By embedding structured metadata within DOI records, the proposed system allows extension specialists and decision-support developers to programmatically identify relevant studies, compare results across environments, and aggregate findings using computational workflows. This approach builds on emerging concepts such as FAIR Digital Objects and PID Graphs (Schwardmann, 2020), while extending them to support domain-specific scientific synthesis. Because DOI metadata exists independently of the underlying publication or dataset, this approach adds structured context without modifying original research outputs or introducing copyright concerns. In practical terms, this capability supports improved recommendations for crop management, pest control, and environmental stewardship by allowing evidence to be updated more rapidly as new studies become available, supporting producers in making more timely, evidence-based management decisions.

The framework consists of two integrated components. The first focuses on the representation of scientific findings, using Crossref-linked metadata to encode key elements of published studies (e.g., variables, measured outcomes, and relationships) into structured, machine-readable formats. This builds on prior work demonstrating the value of structured metadata and computational extraction of scientific knowledge (Molik et al., 2021). The second focuses on the representation of data structure, leveraging and extending DataCite metadata schemas to describe experimental design, variable definitions, units,

and data organization. This component is informed by ongoing efforts to standardize biological metadata and improve interoperability across datasets (Wright et al., 2024).

This approach differs fundamentally from existing efforts in agricultural data management. FAIR and ontology-based initiatives typically operate at the repository level or require post hoc standardization. In contrast, this project embeds interoperability directly into research outputs, enabling integration at the point of publication. By aligning DOI metadata with domain ontologies while maintaining compatibility with Crossref and DataCite systems, the framework leverages existing infrastructure while extending its functionality. This positions agricultural research within emerging efforts around PID Graphs and FAIR Digital Objects, while advancing a practical implementation for scientific synthesis.

The research program will be organized around three integrated activities. First, we will design and implement the dual-layer metadata architecture, developing extensible schemas and templates applicable across diverse agricultural research domains. This work will prioritize parsimony and usability to support adoption within existing publication workflows.

Second, we will implement and refine the framework using representative datasets spanning environmental measurements, crop performance trials, and GGB studies. This will involve selecting published studies, generating PID-linked metadata records encoding findings and data structure, and integrating these records into DOI workflows. Implementation will occur within operational publishing and dissemination infrastructure at Kansas State University, including New Prairie Press and the Open Publishing Exchange. These resources provide a real-world environment for embedding metadata into publication processes and testing how machine-actionable metadata can be generated at the point of publication and linked to underlying datasets.

Third, we will evaluate the impact of PID-embedded metadata on scientific synthesis by comparing traditional meta-analysis with computational workflows enabled by the proposed system. This evaluation will focus on use cases relevant to U.S. agriculture, such as synthesizing multi-location crop trials or integrating pest response data across environments. Metrics will include integration time, reproducibility, scalability, and sensitivity to data heterogeneity, ensuring that the resulting synthesis approaches meet standards for rigorous, reproducible (‘gold standard’) scientific analysis.

The anticipated outcomes extend beyond improved data management to fundamentally enhanced research capability. By enabling cross-study integration at scale, the proposed framework will support more robust agroecosystem models and accelerate the translation of research findings into actionable knowledge. By enabling synthesis across hundreds to thousands of experiments, this work directly supports improved recommendations for crop management, pest control, and climate-resilient agricultural systems.

Key risks include adoption barriers, data heterogeneity, and metadata quality. These will be mitigated through minimal schema design, iterative testing across diverse datasets, and validation workflows. Even if full automation is not achieved, the project will produce interoperable metadata frameworks and empirical insights into enabling computational synthesis.

This project is designed as a focused, high-impact effort carried out by a small research team centered on a dedicated postdoctoral researcher. By leveraging existing PID infrastructure and publishing systems, it minimizes the need for new infrastructure while maximizing impact and feasibility.

In accordance with the RFA, this proposal focuses on the overarching research program rather than a detailed experimental plan. If successful, this work will transform how agricultural knowledge is represented and integrated by redefining DOIs and PIDs as computationally actionable units of scientific information, increasing the return on investment in agricultural research in the United States.

Innovation and Feasibility

State clearly and concisely what makes your project exceptionally innovative in the context of the current state-of-the-art of the field. *

Describe how this research has not been done elsewhere or how it accelerates a current research challenge or addresses it in a new or innovative manner.

If applicable, describe how the proposed research will move the technology, tool, practice or approach under study from its current status further along the commercialization pipeline. Since the approaches may be highly innovative, and hence risky, what will you do if these approaches are not successful? (up to 500 words)

Persistent Identifiers (PIDs), including Digital Object Identifiers (DOIs) are a foundational element of scientific output. The project innovates this principle by turning DOIs from passive reference mechanisms into active, machine-actionable units of scientific knowledge. While PIDs are widely used to link publications, datasets, and research outputs, their current function is primarily relational, enabling discovery but not computation. At the same time, agricultural data initiatives built around FAIR principles have improved accessibility but have not resolved a central limitation: most research outputs remain non-computable at the level of scientific meaning and data structure, preventing scalable integration and synthesis.

This extends PID infrastructure into a computable metadata layer that encodes both scientific findings and data structure directly within DOI-associated records. It introduces a dual-layer architecture that integrates both structured representations of scientific claims through Crossref-linked metadata and formal descriptions of data schema and experimental design through DataCite-linked metadata. While existing efforts address semantic or structural interoperability independently, this project unifies them at the PID layer, aligning with emerging concepts such as PID Graphs and FAIR Digital Objects, while extending them into an operational framework for agricultural research.

A key innovation is the transformation of scientific synthesis into a computational workflow. Rather than relying on manual meta-analysis, the proposed system enables automated querying and aggregation across studies. By encoding variables, relationships, and data structures within PID-linked metadata, PIDs become computational entry points for cross-study analysis, enabling synthesis at a scale and speed not currently possible. Despite its conceptual ambition, the project is highly feasible because it builds on existing, widely adopted PID infrastructure rather than requiring new platforms. Crossref and DataCite already provide extensible metadata systems and are evolving toward interconnected PID ecosystems. The project leverages these systems, minimizing technical barriers while aligning with ongoing international efforts.

Implementation will be carried out by a focused team centered on a postdoctoral researcher, supported by targeted computational resources and collaborations with New Prairie Press and Open Publishing Ecosystem partners. These collaborations provide direct access to publication workflows, enabling rapid prototyping and iterative refinement.

Risks include adoption barriers, data heterogeneity, and metadata quality. These will be mitigated through minimal, modular schema design, iterative testing across diverse datasets, and validation workflows. If full automation is not achieved, the project will still produce interoperable metadata frameworks and empirical insights into enabling computational synthesis.

Investigator Qualifications

Provide evidence to support your claim of innovativeness and creativity in your prior research. For example, what personal qualities and experiences demonstrate your inclination to question paradigms and work with intellectual uncertainties, develop unique collaborations, integrate diverse sources of information, or develop novel approaches when new challenges or opportunities arise? (up to 500 words) *

My research program focuses on addressing structural limitations in how agricultural knowledge is generated, organized, and reused, particularly at the intersection of agricultural systems, data science, and research infrastructure. Rather than working within a single disciplinary framework, I have focused on identifying persistent bottlenecks that limit the impact of agricultural research, especially data fragmentation, inconsistent metadata, and the difficulty of synthesizing results across studies. My research also focuses on developing approaches that connect data, methods, and knowledge across domains and scales.

A central theme of my work has been the integration of heterogeneous agricultural data systems, including environmental observations, crop and pest dynamics, and genomics, genetics, and breeding (GGB) data. In practice, this has required working across differences in data standards, formats, and research cultures, as well as engaging with how data are generated, published, and reused. Through this experience, I have developed an approach that prioritizes translation across systems, technically and across disciplinary and institutional boundaries. This translation enables more effective synthesis and application of research findings.

My interest in this problem is grounded in direct experience with the limitations of current research workflows. Much of agricultural data science remains constrained by project-level silos and retrospective integration, where data are only harmonized after publication, if at all. Through my work with publishing and data infrastructure systems, especially within libraries, I have seen how these limitations arise from technical challenges and from how research outputs are structured and disseminated. This has led me to focus on a different approach: designing low-cost decentralized infrastructure that enables interoperability and synthesis at the point of publication, rather than after.

I have demonstrated the ability to develop and lead interdisciplinary collaborations that bring together expertise from agronomy, data science, informatics, and publishing systems. My affiliation with the ID3A provides access to researchers generating complex, multi-domain datasets, while my leadership in publishing and metadata infrastructure at Kansas State University provides a complementary perspective on how research outputs are structured and disseminated. My leadership within the i5k Initiative and AgBioData further supports these claims. This combination has allowed me to work at the interface between data generation, publication, and reuse, and to identify opportunities for integration not visible from within a single domain.

My work reflects a consistent pattern of engaging with emerging challenges by developing new approaches rather than incrementally extending existing ones. The rapid growth in the scale and complexity of agricultural data, combined with evolving digital infrastructure, creates an opportunity to rethink how scientific knowledge is represented, connected, and published. The proposed research, extending PID systems into machine-actionable scientific infrastructure, builds directly on this trajectory, integrating technical innovation and systems-level thinking.

I am committed to advancing open science, reproducibility, and long-term research impact. Through my work with open publishing platforms and data infrastructure, I have focused on ensuring that research outputs are not only accessible, but also reusable and integrable across studies, aligning with FFAR's goals of increasing the return on investment in agricultural research and accelerating the translation of research into practice.

Suitability for the New Innovator Award program

Why is the planned program fundamental to building your research career? (up to 500 words)*

The proposed research program is fundamental to establishing my long-term research trajectory at the intersection of agricultural systems, data science, and scientific infrastructure. My goal is to develop a sustained program focused on computationally enabled agricultural knowledge systems, where data, publications, and analytical methods are seamlessly integrated to support discovery and decision-making. This project represents a critical inflection point in that trajectory, providing the resources and flexibility needed to transition from conceptual and exploratory work into a cohesive, high-impact research program.

As an early-career investigator, I am at a stage where defining a clear and independent research direction is essential. The New Innovator Award is uniquely suited to support this transition because it prioritizes creative, high-risk ideas that may not yet fit within traditional funding mechanisms. The proposed work, reframing DOI/PID systems as machine-actionable scientific infrastructure, challenges conventional approaches to data management and synthesis in agriculture. Pursuing this direction requires the ability to explore new conceptual frameworks and build foundational tools without the constraints of narrowly defined, incremental projects. This award provides the opportunity to do so in a focused and sustained manner.

This program will enable me to establish a distinct research niche that bridges agricultural science, data infrastructure, and open knowledge systems. While these areas are often treated separately, their integration is essential for addressing the growing complexity of agricultural data and the need for scalable synthesis. By developing methods that embed interoperability and computability directly into research outputs, this work positions me at the forefront of an emerging area gaining attention across scientific domains but remains underdeveloped in agriculture. Establishing leadership in this space will create a strong foundation for future collaborations, funding opportunities, and contributions to national data initiatives.

The structure of the proposed project (a dedicated postdoctoral researcher and partnerships with publishing and data infrastructure organizations) will also support the development of a scalable and sustainable research program. It will allow me to build technical capacity, refine methods, and generate preliminary results that can support larger, multi-institutional efforts in the future. In particular, integration with ID3A provides a pathway for expanding this work into broader networks focused on digital agriculture, data interoperability, and decision-support systems.

Equally important, this program will contribute to my development as a mentor and leader in interdisciplinary research. By training a postdoctoral researcher at the interface of agriculture and data science, I will begin to build a research group that is capable of addressing complex, cross-cutting challenges. This aligns with FFAR's goal of supporting investigators who will train the next generation of scientists equipped to work across disciplinary boundaries.

Finally, the proposed work aligns closely with FFAR's emphasis on long-term impact and the advancement of the U.S. food and agriculture system. By focusing on how research outputs are structured, connected, and reused, this program addresses a foundational challenge that affects the efficiency and effectiveness of agricultural science as a whole. Establishing this line of research now is essential for positioning my career to contribute meaningfully to the future of data-driven agriculture.

Project Data Management and Dissemination Plan

FFAR requires that, where possible, project data and results be shared broadly. To facilitate data and results sharing, a proposal submitted to FFAR must include a Data and Results Management Plan which should include a communication plan to amplify outcomes to stakeholder audiences. Plans should include a description of what type of data are expected, how those data will be preserved and stored; and how data and results will be accessed and shared in a timely manner. Even in the unlikely case in which no data or informational results will be produced, a plan must be submitted which states, "No data are expected to be produced." However, overarching results are still expected to be shared (up to 500 words). *

This project will generate and curate machine-actionable metadata, computational workflows, and synthesized datasets designed to enable cross-study integration of agricultural research outputs. Data management practices will align with FAIR principles while explicitly advancing computability, interoperability, and AI/ML readiness through the use of Persistent Identifiers (PIDs) and structured metadata frameworks.

Data Types and Formats:
The project will produce three primary categories of outputs:
(1) PID-linked metadata records encoding structured representations of scientific findings and data schemas;
(2) curated datasets derived from published agricultural studies (e.g., multi-location field trials, environmental observations, and genomics-enabled datasets)
(3) computational workflows and software used for metadata generation, integration, and synthesis.

All data will be stored in open, non-proprietary, machine-readable formats (e.g., JSON).

Metadata and Persistent Identifiers (PIDs):
A central objective of this project is the development and implementation of enhanced PID-linked metadata frameworks. All datasets, software, and research outputs will be assigned DOIs via DataCite and linked to associated publications through Crossref, forming an interconnected PID graph.

Metadata will extend established standards to include machine-actionable representations of scientific findings and data structure, enabling direct computational use. Documentation (README files, data dictionaries, and workflow descriptions) will ensure interpretability, reproducibility, and reuse.

Storage, Backup, and Security:
Active data will be stored using K-State infrastructure, including institutional storage systems, the Beocat High-Performance Computing cluster, and AWS cloud environments. Data will be backed up regularly using redundant systems, with oversight by the PI. Standard institutional practices will ensure data integrity and security.

Data Sharing and Dissemination:
All outputs will be made openly available in a timely manner, typically at publication or within six months of generation. Data and metadata will be deposited in trusted repositories such as Zenodo, or K-State's K-REx, ensuring persistent access and DOI assignment.

Dissemination will leverage New Prairie Press (NPP) and the Open Publishing Exchange (OPE), enabling data-linked publications and integration of machine-actionable metadata into the scholarly record. Software and workflows will be shared via GitHub, with DOI assignment through Zenodo and long-term preservation via archival systems. Workflows will be containerized where appropriate to support reproducibility.

Key Deliverables:
This project will produce openly available, DOI-registered outputs including:
- A dual-layer PID-linked metadata framework for encoding scientific findings and data structure
- Schema extensions and implementation templates compatible with Crossref and DataCite workflows
- Curated, machine-actionable datasets from representative agricultural studies
- Open-source computational workflows for metadata generation and cross-study synthesis
- Demonstration case studies comparing traditional and computational meta-analysis approaches
- Data-linked publications and technical documentation disseminated through New Prairie Press and partner platforms

Roles and Responsibilities:
The PI will oversee all aspects of data management and compliance. A dedicated postdoctoral researcher will lead metadata development, data curation, and workflow implementation, with support from OPE Data Librarian.

Long-term Preservation and Impact:
All outputs will be preserved for a minimum of 10 years. By structuring research outputs as PID-linked, machine-actionable objects, this project enables direct computational reuse, large-scale synthesis, and integration into modeling and decision-support systems, contributing to a more connected and interoperable agricultural research ecosystem.

Facilities, Equipment and other Resources

Describe the resources available to the research team (up to 500 words).*

Kansas State University (K-State) provides a highly integrated research environment that is fully sufficient to support the proposed project, combining computational infrastructure, data systems, and publishing platforms that are directly aligned with the development of machine-actionable, PID-linked research outputs. Because the project focuses on metadata frameworks and computational workflows it does not require specialized laboratory facilities or major equipment investments. Instead, it leverages existing institutional resources to enable efficient, scalable, and low-risk implementation.

Computational Infrastructure

The project will utilize K-State's Beocat High-Performance Computing (HPC) cluster and institutionally supported Amazon Web Services (AWS) cloud infrastructure. These systems provide flexible and scalable capacity for data processing, metadata generation, and cross-study integration workflows. The hybrid HPC–cloud environment supports both development and deployment of computational pipelines, enabling rapid iteration and testing across datasets of varying size and complexity.

PID, Data, and Publishing Infrastructure

A key strength of the institutional environment is the close integration of persistent identifier systems, repositories, and publishing platforms, which are central to the proposed research. The project will leverage:

- DataCite and Crossref infrastructure for DOI registration, metadata extension, and PID graph development
- K-REx (Kansas State Research Exchange) and trusted external repositories (e.g., Zenodo, Dryad) for long-term data storage and dissemination
- Established workflows for linking datasets, publications, and software through interoperable PID systems

The Open Publishing Exchange (OPE) within K-State Libraries provides specialized expertise in metadata standards, FAIR data practices, and PID implementation. OPE will play a critical role in supporting schema development, repository integration, and alignment with evolving standards for machine-actionable research outputs.

The project will also utilize New Prairie Press (NPP), K-State's open-access publishing platform, which enables the publication of data-linked and metadata-rich research outputs. NPP's integration with DOI systems provides a practical pathway for embedding enhanced metadata into real publication workflows, allowing the proposed framework to be tested and refined within an operational publishing environment.

Software and Reproducibility Resources

All computational work will be conducted using widely adopted, open-source tools for reproducible research. Code and workflows will be developed and shared via GitHub, with DOI assignment through Zenodo integration and long-term preservation through archival systems. Containerization technologies (e.g., Docker or Singularity) will be used where appropriate to ensure portability and reproducibility across computational environments.

Collaborative and Interdisciplinary Environment

The project is further strengthened by its connection to the Institute for Digital Agriculture and Advanced

Analytics (ID3A) at K-State, which provides an interdisciplinary environment linking agricultural science, data analytics, and computational research. ID3A facilitates collaboration with researchers generating diverse datasets across crop systems, environmental monitoring, and genomics, providing essential real-world contexts for testing and validating the proposed framework.

Summary

Together, these resources provide a cohesive, end-to-end infrastructure for developing, implementing, and disseminating machine-actionable, PID-linked research outputs. By leveraging existing computational, data, and publishing systems the project is both cost-effective and highly feasible, while uniquely positioned to translate conceptual advances into practical, widely adoptable solutions for agricultural research.

Budget Details

Instructions

Match is optional for this funding opportunity.
Budget requests should be developed commensurate with the support needed to achieve the project goals. You may request up to US $450,000.00 from FFAR, contingent on funding availability.

Please refer to FFAR's Applicant Forms & Examples website for budget templates.

- You may request up to $150,000/year for up to three years (36 months).
- If your project includes one or no sub-awards, please use the Single Sub-Award Budget Template.
- If your project includes more than one sub-award, please use the Multiple Sub-Awards Budget Template.
- Budget requests should be developed commensurate with the support needed to achieve the project goals. A completed budget spreadsheet will have to be uploaded to the Attachments section before submission.
- Amounts provided in this section should correspond to the totals in your FFAR Budget Template Excel file. If the Grantee is selected for funding and a discrepancy is found between the FFAR Budget Template and this section, FFAR will defer to the values entered below as the official budget being requested. Please note that all totals should include direct and indirect costs.
- If you do not have cumulative projected costs in a specific category, please enter "0" (zero).

Budget Justification

Justification*

Personnel
Year One: $74,678.00, Year Two: $76,918.00 Year Three: $79,226.00 Total $230,822.00

Funds are requested to support one Postdoctoral Research Associate who will be responsible for the technical implementation of the project. The postdoctoral researcher will lead the development of data integration workflows, metadata structuring, and machine-actionable infrastructure, including the extension of Persistent Identifier (PID) systems.

In addition to technical responsibilities, this position includes a structured mentorship component. The Postdoctoral Research Associate will receive training in interdisciplinary research at the intersection of agricultural systems, data science, and research infrastructure, including experience in collaborative project design, stakeholder engagement, and open science dissemination. The PI will provide regular guidance to support the development of both technical expertise and independent research leadership.

This role is central to project success, enabling rapid prototyping, testing, and deployment of project outputs. Salary levels are aligned with local federal postdoctoral compensation benchmarks.

Fringe Benefits
Year One: $25,391.00, Year Two: $26,152.00, Year Three: $26,937.00, Total: $78,480.00

Fringe benefits are calculated in accordance with Kansas State University's federally negotiated rates and include standard employee benefits such as health insurance and retirement contributions.

Materials and Supplies (Computers and Monitors)
Year One: $7,300, Year Two: 3,000, Year Three: $0, Total: 10,300

Funds are requested for computing equipment, including a workstation and monitors, to support the Postdoctoral Research Associate. These resources are necessary to develop, test, and manage data-intensive workflows, including handling large, heterogeneous agricultural datasets and supporting local development environments.

The requested equipment provides appropriately configured computational resources to support efficient project implementation.

Computer Services (AWS and Beocat)
Year One: $12,000, Year Two: $12,360, Year Three: 12,361, Total: $36,721

Funds are requested for cloud and high-performance computing services, including AWS and Kansas State University's Beocat cluster.

These resources will support:

- Scalable data storage and processing
- Development and deployment of data infrastructure
- Testing of workflows across distributed and high-performance environments

Cloud and HPC resources are necessary for working with large-scale agricultural datasets and support flexible scaling aligned with project needs.

Publication / Documentation / Dissemination (APCs)
Year One: $5,000, Year Two: $5,150, Year Three: $5,305, Total: $15,455

Funds are requested for Article Processing Charges (APCs) to support open access publication of project results.

Open dissemination ensures that project outputs are accessible and reusable across the agricultural research community. By providing access to methods, data structures, and infrastructure approaches, these publications support broader adoption across research and applied contexts.

Travel (Domestic)
Year One: $6,000.00, Year Two: $6,850.00, Year Three: $6,600.00, Total: $19,450.00

Domestic travel funds will support participation in scientific meetings, workshops, and stakeholder engagement activities.

Travel will enable:

- Dissemination of project findings
- Engagement with agricultural researchers and data practitioners
- Development of collaborations that support adoption and long-term use

These activities support alignment of project outputs with community needs.

Indirect Costs
Year One: $19,555.35, Year Two: $19,564.50, Year Three: $19,564.35, Total: $58,684.20

Indirect costs are calculated in accordance with Kansas State University’s federally negotiated rate agreement and support institutional infrastructure, administrative services, and compliance.

Overall Budget Rationale

The budget prioritizes personnel and computational resources necessary for the development and implementation of interoperable agricultural data systems. Resources are allocated to support data infrastructure development, dissemination, and engagement activities that facilitate adoption within the agricultural research community.

This approach supports both technical development and integration into existing research workflows. The budget is tightly scoped to support a single integrated research effort centered on personnel and scalable infrastructure, minimizing overhead while maximizing impact.

The fringe benefit rates are determined by the State of Kansas Department of Administration and communicated through K-State's Division of Human Resources. The fringe rates for post doc salary have been calculated at 34%.
https://www.k-state.edu/research/faculty/proposal/budgets/fringe-rates.html

Indirect Costs : Indirect costs are calculated in accordance with Kansas State University’s federally negotiated rate agreement and support institutional infrastructure, administrative services, and compliance. For this project the indirect costs are calculated at sponsor allowed 15% total direct costs.

Cumulative Budget

If you do not have projected costs in a specific category, please enter "0" (zero).

Total Budget Request*
FFAR + Match. The total amount must account for direct and indirect costs.

449,912.2

Total FFAR Funds Requested*

449,912.2

Total Match
Cash Match + In-Kind Match

0

Total Cash Match

0

Total In-Kind Match

0

Confirm Total Budget Request Amount*

$449,912.20 USD

Year 1 Budget Request

Year 1 Total Budget*

149,924.35

FFAR Funds Requested*

149,924.35

Total Match

0

Cash Match

0

In-Kind Match

0

Year 2 Budget Request

Year 2 Total Budget
If not applicable, please leave blank.

149,994.5

FFAR Funds Requested

149,994.5

Total Match

0

Cash Match

0

In-Kind Match

0

Year 3 Budget Request

Year 3 Total Budget
If not applicable, please leave blank.

149,993.35

FFAR Funds Requested

149,993.35

Total Match

0

Cash Match

0

In-Kind Match

0

Subaward 1

Subaward Name

No answer

Institution Type

No answer

Institution Profit Status Type

No answer

Subaward Tax ID

No answer

Subaward State

No answer

Subaward Country

No answer

Subaward Total

No answer

Subaward FFAR Funds

No answer

Subaward Total Match

No answer

Subaward Cash Match

No answer

Subaward In-Kind Match

No answer

Subaward 2

Name

No answer

Institution Type

No answer

Institution Profit Status Type

No answer

Tax ID

No answer

State

No answer

Country

No answer

Subaward Total

No answer

FFAR Funds

No answer

Total Match

No answer

Cash Match

No answer

In-Kind Match

No answer

Subaward 3

Name

No answer

Institution Type

No answer

Institution Profit Status Type

No answer

Tax ID

No answer

State

No answer

Country

No answer

Subaward Total

No answer

FFAR Funds

No answer

Total Match

No answer

Cash Match

No answer

In-Kind Match

No answer

Match Information

Instructions

Match is optional for this funding opportunity. You may leave this tab blank if you do not have matching funds. For more information about our match policy, please refer to our Matching Guidelines page.

Matching Funds Distribution

Is any of the match listed below being sent directly to FFAR for distribution?

No

Matching Partner (1)

Matching Contributor Name

No answer

Matching Contributor Tax ID

No answer

Matching Contributor Profit Status

No answer

Matching Contributor Type

No answer

Is the match being sent directly to FFAR for distribution?

No answer

Matching Contributor State

No answer

Country (only if outside of the U.S.)

No answer

Total Match

No answer

Cash Match

No answer

In-Kind Match

No answer

Matching Partner (2)

Name

No answer

Tax ID

No answer

Profit Status

No answer

Type

No answer

Is the match being sent directly to FFAR for distribution?

No answer

State

No answer

Country (only if outside of the U.S.)

No answer

Cash Match

No answer

In-Kind Match

No answer

Total Match

No answer

Matching Partner (3)

Name

No answer

Tax ID

No answer

Profit Status

No answer

Type

No answer

Is the match being sent directly to FFAR for distribution?

No answer

State

No answer

Country (only if outside of the U.S.)

No answer

Cash Match

No answer

In-Kind Match

No answer

Total Match

No answer

Matching Partner (4)

Name

No answer

Tax ID

No answer

Profit Status

No answer

Type

No answer

Is the match being sent directly to FFAR for distribution?

No answer

State

No answer

Country (only if outside of the U.S.)

No answer

Cash Match

No answer

In-Kind Match

No answer

Total Match

No answer

Matching Partner (5)

Name

No answer

Tax ID

No answer

Profit Status

No answer

Type

No answer

State

No answer

Is the match being sent directly to FFAR for distribution?

No answer

Total Match

No answer

Cash Match

No answer

Country (only if outside of the U.S.)

No answer

In-Kind Match

No answer

Organizational Assurances

Please select all that apply.*

None

IRB

All projects involving human subjects must have approval from an institutional review board (IRB), unless they qualify for an exemption from IRB review, before an award can be made. Where IRB approval is required, applicants should file their proposal with their local IRB at the same time the proposal is submitted to the Foundation for Food and Agriculture Research, so that the approval procedure will not delay the award process. A proposal may be submitted to the Foundation for Food and Agriculture Research without IRB approval; but if the proposal is selected for funding, award will be made conditional upon IRB approval within 60 days of the notice of conditional award. If a proposed project involving human subjects is exempt from human subjects regulations [see 45 CFR 46.101(b)], the applicant must provide documentation that an IRB (or some authority other than the project director or key personnel) has declared the project exempt from the human subjects regulations. Documentation should include the specific category justifying the exemption.

Are human subjects involved in the proposed research?*

No

IRB Status

No answer

IRB Exemption Category

No answer

IRB Submission Date	IRB Approval Date	IRB Assurance Number
No answer	No answer	No answer

IACUC

All projects involving animal use must have approval from an institutional animal care and use committee (IACUC), unless they qualify for an exemption from IACUC review, before an award can be made. Where IACUC approval is required, applicants should file their proposal with their local IACUC at the same time the proposal is submitted to the Foundation for Food and Agriculture Research, so that the approval procedure will not delay the award process. A proposal may be submitted to the Foundation for Food and Agriculture Research without IACUC approval; but if the proposal is selected for funding, award will be made conditional upon IACUC approval within 60 days of the notice of conditional award. If a proposed project involving animal use is exempt from animal use regulations [see The Animal Welfare Act (AWA) (7 U.S.C. § 2131)], the applicant must provide documentation that an IACUC (or some authority other than the project director or key personnel) has declared the project exempt from the animal use regulations. Documentation should include the specific category justifying the exemption.

Are vertebrate animals involved in the proposed research?*

No

IACUC Status

No answer

The following are the only three exemption categories allowed. If an exemption is granted, please type the category in the field below.

- **Activities involving animals that perform tasks, participate in club activities, or appear in exhibits or demonstrations.**
- **Use of tissues, organs or other parts of dead animals if received as such.**
- **Noninvasive observation of wild animals in their natural habitat.**

No answer

IACUC Submission Date	IACUC Approval Date	IACUC Assurance Number
No answer	No answer	No answer

rDNA

When applicable, Research on recombinant DNA (rDNA) should be conducted in accordance with CDC/NIH Biosafety Guidelines (Biosafety in Microbiological and Biomedical Laboratories) and NIH Guidelines for Research Involving Recombinant DNA.

Is Recombinant DNA (rDNA) involved in the proposed research?*

No

rDNA Status

No answer

rDNA Exemption Category

No answer

rDNA Submission Date

No answer

rDNA Approval Date

No answer

rDNA Assurance Number

No answer

National Security

Does the proposed project require review for activities that have possible national security implications? *

No

If "Yes," please provide detailed explanation.

No answer

Hazardous Materials

Are hazardous materials used or produced in the proposed project?*

No

If "Yes," please provide detailed explanation.

No answer

Human Fetal Tissue

Is human fetal tissue used in the proposed project?*

No

If "Yes," please provide detailed explanation.

No answer

NEPA

Does the proposed work require a NEPA review?*

Projects that will likely have a significant impact or anticipated individual cumulative effect on the quality of the environment will typically require preparation of an Environmental Assessment or an Environmental Impact Statement, unless categorically excluded from a detailed environmental analysis. Documentation should include the specific category justifying the exclusion. Projects pending NEPA approval cannot start until NEPA approval is obtained.

No

If "Yes," please provide detailed explanation.

No answer

Attachments

Additional Materials
Please upload attachments to your proposal below. Direct questions or concerns to the FFAR Grants Team at grants@foundationfar.org

Required Attachments

Budget Spreadsheet (Excel)

- If your project includes one or fewer sub-awards, please use the Single Sub-Award Budget Template.
- If your project includes more than one sub-award, please use the Multiple Sub-Awards Budget Template.
- Budget requests should be developed commensurate with the support needed to achieve the project goals.
- FFAR's attachment template must be used. It is available for download at FFAR's Applicant Forms & Examples website.

Current and Pending Support (PDF)

- Please upload a PDF that lists all ongoing and upcoming projects for the past 3 years for the PI/PD and any key personnel named in the application. Please combine all support into a single PDF document before uploading as an attachment.
- FFAR's attachment template must be used. It is available for download at FFAR's Applicant Forms & Examples website.

Project Timeline (PDF)
Applicants must provide a PDF project timetable that outlines anticipated milestones to be tracked for each year of the project. The timeline will be used to evaluate annual progress reports, and is a criterion for continued support of successful applications. If applicable, include periods beyond the grant funding to demonstrate the impact and longevity of the work.

References Cited (PDF)
Applicants must upload a PDF document of their references cited in their Project Description. References Cited do not count towards the word limit.

Biosketches (PDF)
A biographical sketch/CV (up to 5-pages per individual) is required for the PI and all Key Personnel on the proposed project. Please combine all biosketches into a single PDF document before uploading as an attachment.
Do not submit personal information such as home address; phone numbers; home e-mail address; date of birth; citizenship; drivers' license, social security, or passport numbers; marital status; personal hobbies; and the like. Such personal information is irrelevant to the merits of the proposal. Biosketches will be shared with internal and external reviewers and if such information is included, the Foundation for Food & Agriculture Research (FFAR) is not responsible or in any way liable for keeping the information confidential

Optional Attachments

Matching Funds Contribution Letters (PDF)
Please refer to the FFAR website for FFAR's Matching Guidelines and the Matching Funds Contribution Letter Template. Upload a single PDF that contains all of the required Matching Funds Contribution letter(s) for each of the Matching Partner(s) listed in this application. Please combine all letters into a single PDF document before uploading as an attachment.

Additional Key Personnel (PDF)
Applicants with more than three (3) key personnel must upload a table including the name, institution, expertise, role, email, and phone number of any additional key personnel in addition to the three (3) individuals listed on the Contact Information section of the online application.

Graphics, Figures, Equations, and Tables (up to 5-pages) (PDF)
The online application does not support equations, tables, graphics, and figures. Applicants may upload a PDF document with any of these to support the research program plan.

Letters of Support (PDF)
Applicants may provide letters of institutional, collaborator, or stakeholder support for the proposed project. Please combine all letters of support into a single PDF document before uploading as an attachment.

Budget Spreadsheet*

See FFAR_Budget_Worksheet.xlsx

Current and Pending Support*

See FFAR Current and Pending.pdf

Project Timeline*

See FFAR_Project_Timeline.pdf

References Cited*

See FFAR References Cited.pdf

Biosketches*

See FFAR Biosketch - Molik.pdf

Matching Funds Contribution Letters

No file uploaded

Additional Key Personnel

No file uploaded

Graphics and Tables

No file uploaded

Letters of Support

No file uploaded

Additional Attachment

No file uploaded